



RODNEY'S ROVER

STUDY GUIDE

INGENIUM

INTRODUCTION

Delegates will engage in a series of robotics and engineering challenges designed to test their technical skills, creativity, and teamwork. Across multiple rounds, they will code, build, and strategize. Performance in each stage will determine their overall standing in the competition.

ROUND 1 - ROBOT INSPECTION

Participants are required to bring pre-built robots to this round and ensure that their robots comply with the technical and operational guidelines.

Robots will be inspected by the judges to ensure that they meet the required criteria. Once approved, teams will be allowed to practice on the provided track.

Teams will also have the opportunity to build their robots at school for an additional cost.

Note: Teams that choose to build their robots at school must inform the category heads. Delegates must only bring the components required for building their robots. Ingenium will provide only the basic tools required for assembly.

Time allowed: 3 Hours

ROUND 2 - LINE FOLLOWING (ELIMINATION ROUND)

The best completion time for each team will be recorded on the given line-following track.

Each team will be invited to run their robot on the track, and their fastest completion time will be recorded. The top three teams will receive a hidden advantage in the next round.

Delegate cap: 3-5 delegates

Time allowed: 2 Hours



ROUND 3 - SUMO WARS

The final showdown where the delegates will battle it out for glory. The exact bracket format will be announced at a later date.

Delegate cap: 2-3 delegates

Time allowed: 2 Hours

Match Duration: 10 minutes per match

Materials allowed

1. Arduino-based microcontroller.
2. Any motor driver except the IBT-2 motor driver.
3. Any geared DC motor, except high-end series such as the 775.
4. Caster wheel (optional).
5. Plywood or acrylic chassis (custom-built chassis are allowed; metal and 3D-printed chassis are not allowed).
6. Li-Po batteries are not allowed. Only one battery pack may be used to power the entire system, with a maximum voltage of 13V.
7. Any IR sensor, such as LM393, BFD1000, QTR-8, or TCRT5000.
8. HC-05 Bluetooth module.
9. Wires of all types (male-to-male, female-to-female, and male-to-female).

ACTUATOR FOR SUMO (OPTIONAL)

Participants may design and use an actuator for their sumo robot.

- **Small motors:** Maximum of 2, such as MG90-series motors.
- **Heavy motors:** Maximum of 1, such as the MG995R.

ROBOT DIMENSIONS AND WEIGHT

- **Maximum Dimensions:** 1 ft × 1 ft
- **Maximum Weight:** 2.5 kg

LIFTING MECHANISM

A lifting mechanism is allowed, subject to the rules specified in the rulebook.



Rules:

Allowed

- Use only the provided kit components given in the list.
- Teams may write/upload their own Arduino code.
- Teams must bring their own laptop + Arduino IDE for coding.
- Teams are allowed to use lifting/linear push mechanisms for Sumo Wars but mechanisms like pinching/high-velocity strikes are not allowed.
- Recalibration between rounds is allowed.
- Adjustments using tape, glue, or cardboard reinforcements are permitted (non-electrical).

Not Allowed

- External sensors, motors, or custom hardware beyond the specifications of the component list.
- High power external batteries not approved by organizers.
- Any attempt to damage another team's robot intentionally outside the rules of Sumo Wars.
- Battery pack exceeding 13V.
- Dimension of the robot exceeding 1 foot x 1 foot.
- Microcontrollers other than Arduino.